

PGD PRACTICE QUESTIONS — FULL SOLUTIONS

CMS707

Artificial Intelligence



Course: CMS707 — Artificial Intelligence



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Content: Questions 1–11 | Complete Worked Answers

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State Space Search

Agents & Rationality

Predicate Logic

PEAS

Propositional Logic

"Surely computers cannot be intelligent — they can do only what their programmers tell them."

Is the latter statement true? Does it imply the former?

The latter statement is only partially true — and does NOT imply the former.

IS THE "PROGRAMMERS TELL THEM" CLAIM TRUE?

Only partially. ML systems learn from data and generate behaviours their programmers never explicitly coded. AlphaZero was not told how to play chess — it discovered strategies through self-play. At the hardware level, rules are followed — but "being told" $\neq$ "having all behaviour pre-specified."

DOES IT IMPLY COMPUTERS CAN'T BE INTELLIGENT?

No. Humans also "do only what their neurons tell them" — governed by electrochemistry and physics. Yet we consider humans intelligent. The argument, if valid against machines, would also invalidate human intelligence. The mechanism of behaviour $\neq$ the quality of behaviour.

Dr. Orji's framework (AIMA): AI is about building systems that act *rationally* to maximally achieve predefined goals. Rationality is judged by *outcomes and adaptation* — not by whether behaviour originates from biology or code.

Framework: An AI system perceives its environment, acts rationally toward goals, and adapts over time.

SYSTEM	AI LEVEL	REASONING
(i) Barcode Scanners	Minimal	Laser reads pattern → database lookup. No learning, no adaptation, no reasoning. Essentially a function table. Not AI.
(ii) Web Search Engines	High AI	Uses NLP, ML ranking algorithms, personalisation, link analysis (PageRank as inference). Perceives, reasons, adapts. Genuine AI application.
(iii) Voice-Activated Menus	Moderate	Basic = keyword matching (not AI). Advanced = speech recognition + NLP + dialogue management. Extent depends on sophistication level.
(iv) Dynamic Internet Routing	Significant AI	OSPF/BGP perceive network state (congestion, failures), compute optimal paths dynamically, adapt in real-time. Goal-directed, environment-sensitive — matches Dr. Orji's rational agent definition.

Conclusion: Not all computer systems are AI. Key distinguishing factors: *adaptation, goal-directedness, reasoning from percepts, and handling uncertainty.*

A farmer must transport a **Lion (L)**, **Goat (G)**, and **Yam (Y)** across a river. Boat holds farmer + one item. Lion+Goat alone → disaster. Goat+Yam alone → disaster.

STATE REPRESENTATION

State = (F, L, G, Y)
Each value: 0 = left bank, 1 = right bank

INITIAL & GOAL

Start: (0, 0, 0, 0) — all left
Goal: (1, 1, 1, 1) — all right

UNSAFE STATES (INVALID)

$L = G \neq F \rightarrow$ Lion eats Goat
 $G = Y \neq F \rightarrow$ Goat eats Yam

10 VALID STATES

STATE (F,L,G,Y)	SAFE?
(0,0,0,0) Start	✓
(0,0,0,1) Yam right	✓
(0,0,1,0) Goat right	✓
(0,1,0,0) Lion right	✓
(0,1,0,1) L+Y right	✓
(1,0,1,0) F+G right	✓
(1,0,1,1) F+G+Y right	✓
(1,1,0,1) F+L+Y right	✓
(1,1,1,0) F+L+G right	✓
(1,1,1,1) GOAL	✓

Algorithm: BFS (FIFO Queue) — guarantees shortest path. Checks for repeated states is *essential* to prevent infinite cycling.

✓ SOLUTION 1 — 7 STEPS

STEP	ACTION	STATE
Start	—	(0,0,0,0)
1 →	Take Goat	(1,0,1,0)
2 ←	Go alone	(0,0,1,0)
3 →	Take Lion	(1,1,1,0)
4 ←	Bring Goat back	(0,1,0,0)
5 →	Take Yam	(1,1,0,1)
6 ←	Go alone	(0,1,0,1)
7 →	Take Goat	(1,1,1,1) ✓

✓ SOLUTION 2 — 7 STEPS (SWAP L & Y)

STEP	ACTION	STATE
Start	—	(0,0,0,0)
1 →	Take Goat	(1,0,1,0)
2 ←	Go alone	(0,0,1,0)
3 →	Take Yam	(1,0,1,1)
4 ←	Bring Goat back	(0,0,0,1)
5 →	Take Lion	(1,1,0,1)
6 ←	Go alone	(0,1,0,1)
7 →	Take Goat	(1,1,1,1) ✓

Repeated states check — YES, absolutely essential. Without it, the search cycles infinitely: (0,0,0,0)→(1,0,1,0)→(0,0,0,0)→... The reached set terminates BFS in O(10) with only 10 valid states.

- i** **Agent** — Anything that *perceives* its environment through **sensors** and *acts* upon it through **actuators**. Operates automatically, persists over a prolonged time period, adapts to changes, and creates and pursues goals. (Dr. Orji / AIMA)
- ii** **Agent Function** — An *abstract mathematical description* that maps any given percept sequence to an action: $f : P^* \rightarrow A$. It is the theoretical specification of what the agent should do — distinct from the agent program which implements it in code.
- iii** **Rationality** — Doing the *right thing*: selecting actions that **maximise expected performance** according to a defined **performance measure**, given the agent's current knowledge and percept sequence. A rational agent is not omniscient — it acts optimally with the information it has. Dr. Orji: "*a rational agent acts to achieve the best or expected outcome when there is uncertainty.*"
- iv** **Goal-Based Agent** — An agent that has explicit **goals** (descriptions of desirable world states) and chooses actions that lead toward achieving those goals. Unlike simple reflex agents that only react to current percepts, goal-based agents reason about *future states* and *plan action sequences*. More flexible — same goal can be pursued via different paths.

QUESTION 5

PEAS Task Environment Descriptions

PEAS = Performance Measure | Environment | Actuators | Sensors

ACTIVITY	PERFORMANCE	ENVIRONMENT	ACTUATORS	SENSORS
v. Football	Goals scored, possession %, match result	Pitch, 21 players, referee, ball, weather	Legs, head, arms, body	Eyes, ears, proprioception
vi. Online Book Shopping	Correct book, lowest price, good condition, secure checkout	Internet, e-commerce sites, search engines, payment gateways	Fill forms, click links, submit orders	HTML pages, prices, ratings, search results
vii. Tennis Match	Points, games, sets won; errors minimised	Court, opponent, ball, net, umpire, weather	Racket swing, footwork, serve motion	Eyes (ball, opponent), proprioception, sound
viii. Auction Bidding	Win item at minimum price; avoid overbidding	Auction house/online, other bidders, auctioneer, time limit	Bid submission (paddle/button/voice), withdraw	Current bid, time remaining, other bidders, item condition

QUESTIONS 6 & 7

Heuristic Search & State Space Terminology

Heuristic Search (Q6) — Uses problem-specific knowledge encoded in a heuristic function $h(n)$ to estimate the cost from node n to the goal, guiding search toward more promising paths.

DIMENSION	BASIC SEARCH (BFS/DFS)	HEURISTIC SEARCH (A*, BEST-FIRST)
Knowledge	None — blind/uninformed	Uses $h(n)$ to estimate goal distance
Direction	Systematic, no preference	Guided toward promising states
Speed	Slower on large spaces	Significantly faster
A* formula	—	$f(n) = g(n) + h(n)$

STATE SPACE TERMS (Q7)

i. State — Complete description of the world at one moment; all info relevant to the problem.

ii. State Space — All states reachable from the initial state via legal actions. Represented as a graph.

iii. Search Tree — Tree generated during search. Each node = state reached by action sequence. Same state may appear in multiple branches.

iv. Search Node — Data structure holding: state, parent node, action taken, path cost $g(n)$, depth. What the algorithm actually manipulates.

v. Goal — The desired final state the agent tries to reach. Search terminates when goal is found.

vi. Action — Operation that transitions state s to $RESULT(s,a)$. $ACTIONS(s)$ returns all valid actions in state s .

True or False Assertions (i–iv)

FALSE

i. An agent sensing only partial info cannot be perfectly rational.

Rationality is defined relative to *available* information. An agent doing the best it can with partial percepts is rational by definition. Example: a card player cannot see opponents' hands yet can still be perfectly rational by using probability.

TRUE

ii. There exists a task environment in which every agent is rational.

A trivial environment where every action leads to the goal, or all actions produce identical outcomes. Any agent — even random — is rational since every choice is equally optimal.

TRUE

iii. There exist environments where no pure reflex agent can behave rationally.

A pure reflex agent maps only the current percept $\rightarrow$ action with no memory. In partially observable environments where correct action depends on history, it cannot behave rationally. Example: a navigation task requiring memory of visited rooms.

FALSE

iv. The input to an agent program is the same as the input to the agent function.

The **agent function** takes the entire *percept sequence* P^* . The **agent program** receives only the *current percept* as input and maintains internal state to simulate the history. Different inputs.

FALSE v. Every agent function is implementable by some program/machine combination.

Functions requiring solving undecidable problems (halting problem), infinite memory, or infinite computation per step cannot be implemented. The set of computable functions is a strict subset of all possible mappings $P^* \rightarrow A$.

TRUE vi. A random-action agent can be rational in some deterministic task environment.

Consider an environment where only one state exists (initial = goal), or all actions are equivalent. Any action is equally good, so random selection is as rational as any other strategy.

TRUE vii. An agent can be perfectly rational in two distinct task environments.

If two environments have compatible performance measures and the same optimal behaviour, the agent is rational in both. Example: a shortest-path agent is perfectly rational both in city-route planning and maze-solving — both reward minimising path cost.

FALSE viii. Every agent is rational in an observable environment.

Full observability means the agent can see the complete state — but does not guarantee correct action choice. A poorly designed agent that ignores its percepts, or always picks the wrong action, is irrational even with full information. Rationality requires having information *and* using it correctly.

i Can multiple agent programs implement the same agent function?

YES. The agent function is an abstract mapping $f: P^* \rightarrow A$. Multiple programs can realise the same mapping. Example: a lookup-table agent and a recursive rule-based agent can produce identical percept-to-action outputs. Two sorting algorithms both implement the abstract "sort" function. Architecture differs; the logical mapping is the same.

ii Are there agent functions that cannot be implemented by any program?

YES. Functions requiring solving undecidable problems (e.g., the halting problem), infinite memory, or infinite computation per step are not implementable by any finite program or machine. The computable functions are a strict subset of all possible $P^* \rightarrow A$ mappings.

iii Given a fixed architecture, does each agent program implement exactly one agent function?

YES (for deterministic programs). A deterministic program on a fixed architecture produces exactly one output per input sequence $\rightarrow$ implements exactly one agent function. If the program uses randomness, it implements a stochastic function (a probability distribution over actions).

iv Speeding up the machine by 2× — does the agent function change?

Generally NO — the abstract percept-to-action mapping is defined by algorithm logic, not speed. **Exception:** In time-bounded environments, faster hardware enables deeper search per move (e.g., chess), effectively producing better decisions — acting as a higher-quality agent function within the time constraint.

Propositional Logic — Which are Correct? (a–e)

$\models$ = entails (every model satisfying LHS also satisfies RHS) | $\equiv$ = logical equivalence

a. $\text{False} \models \text{True}$

✓ CORRECT

Vacuously true — no model satisfies False, so the entailment holds trivially. (*Ex falso quodlibet* — from a contradiction, anything follows.)

b. $\text{True} \models \text{False}$

✗ INCORRECT

True holds in every model. False holds in no model. There are models where True holds but False does not. Entailment fails.

c. $(A \wedge B) \models (A \Leftrightarrow B)$

✓ CORRECT

If $A \wedge B$ is true then $A=T$ and $B=T$. When both true, $A \Leftrightarrow B = T$. Every model satisfying $A \wedge B$ also satisfies $A \Leftrightarrow B$. ✓

d. $A \Leftrightarrow B \models A \vee B$

✗ INCORRECT

Counterexample: $A=F, B=F \rightarrow A \Leftrightarrow B = T$ (both false $\rightarrow$ biconditional true), $A \vee B = F$. LHS true, RHS false. Entailment fails.

e. $A \Leftrightarrow B \models \neg A \vee B$

✓ CORRECT

$\neg A \vee B \equiv A \Rightarrow B$. And $A \Leftrightarrow B = (A \Rightarrow B) \wedge (B \Rightarrow A)$, which logically implies $A \Rightarrow B$. Check: $A=T, B=T \rightarrow$ both T ✓; $A=F, B=F \rightarrow A \Leftrightarrow B=T, \neg A \vee B=T$ ✓. All cases hold.

Propositional Logic — Which are Correct? (f–i)

f. $(A \wedge B) \Rightarrow C \models (A \Rightarrow C) \vee (B \Rightarrow C)$

✓ CORRECT

If $A=F$: $A \Rightarrow C=T \rightarrow \text{RHS}=T$ ✓. If $B=F$: $B \Rightarrow C=T \rightarrow \text{RHS}=T$ ✓. If $A=T, B=T$: LHS forces $C=T$, so both $A \Rightarrow C=T$ and $B \Rightarrow C=T \rightarrow \text{RHS}=T$ ✓. All cases hold.

g. $(C \vee (\neg A \wedge \neg B)) \equiv ((A \Rightarrow C) \wedge (B \Rightarrow C))$

✓ CORRECT

$(A \Rightarrow C) \wedge (B \Rightarrow C) = (\neg A \vee C) \wedge (\neg B \vee C)$. By distribution: $= C \vee (\neg A \wedge \neg B)$. Both sides are identical by De Morgan / distribution. Logical equivalence holds. ✓

h. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B)$

✓ CORRECT

The LHS is a conjunction that contains $(A \vee B)$ as a conjunct. If the full conjunction is true, each conjunct must be true. So $(A \vee B)$ is trivially entailed — a conjunction always entails each of its conjuncts.

i. $(A \vee B) \wedge (\neg C \vee \neg D \vee E) \models (A \vee B) \wedge (\neg D \vee E)$

✗ INCORRECT

Counterexample: $A=T, B=T, C=F, D=T, E=F$.
LHS: $(T) \wedge (T \vee F \vee F) = T \wedge T = \mathbf{T}$.
RHS: $(T) \wedge (F \vee F) = T \wedge \mathbf{F} = \mathbf{F}$. Entailment fails. ✗

Agip Estate Friendship Club — Predicate Logic

Members: Kalu, Ejiro, Sotonye, Uju | Kalu married Ejiro | Sotonye is Uju's brother | Last meeting at Kalu's house

PREDICATE LOGIC REPRESENTATION

Fact 1: $\text{Member}(\text{Kalu}) \wedge \text{Member}(\text{Ejiro})$
 $\wedge \text{Member}(\text{Sotonye}) \wedge \text{Member}(\text{Uju})$

Fact 2: $\text{Married}(\text{Kalu}, \text{Ejiro})$
 $\wedge \text{Married}(\text{Ejiro}, \text{Kalu})$
[marriage is symmetric]

Fact 3: $\text{Brother}(\text{Sotonye}, \text{Uju})$

Fact 4: $\forall x \forall y: \text{Member}(x) \wedge \text{Married}(x, y)$
 $\rightarrow \text{Member}(y)$

Fact 5: $\text{LastMeetingAt}(\text{house}(\text{Kalu}))$

WHAT CAN WE DERIVE?

From Facts 1+4: Since Kalu is a member and married to Ejiro $\rightarrow$ Ejiro must be a member $\checkmark$ (consistent with Fact 1).

From Facts 2+4: Ejiro is married $\rightarrow$ Ejiro's spouse (Kalu) is also a member $\checkmark$.

Sotonye and Uju: only brother relationship given — no marriage stated.

ADDITIONAL STATEMENTS TO PROVE

6. Last meeting at Ejiro's house
7. Uju is not married

Deriving Statements 6 & 7 — Proof

Statement 6 — Last meeting at Ejiro's house

REQUIRES ADDITIONAL FACT (6*)

Facts 1–5 do not explicitly state that married couples share a house.

Add: $\forall x \forall y: \text{Married}(x, y) \rightarrow \text{house}(x) = \text{house}(y)$

- Married(Kalu, Ejiro) — Fact 2
- $\text{house}(\text{Kalu}) = \text{house}(\text{Ejiro})$ — Fact 6* + Fact 2
- LastMeetingAt($\text{house}(\text{Kalu})$) — Fact 5
- ⋮ **LastMeetingAt(house(Ejiro)) ✓**

Statement 7 — Uju is not married

CLOSED WORLD ASSUMPTION (CWA)

Only stated marriage: Kalu–Ejiro. Sotonye is Uju's brother — siblings don't marry. No external marriages mentioned. Under CWA: any fact not provable from KB is assumed false.

$\therefore \neg \exists x: \text{Married}(\text{Uju}, x)$ ✓
Uju is not married.

⚠ TRUTH TABLE LIMITATION

Truth tables are limited to propositional logic. This problem requires predicate logic (quantifiers $\forall, \exists$). For full rigour, a **resolution proof** or **forward-chaining inference** (as taught by Dr. Orji) is the correct mechanism.

SUMMARY — ALL QUESTIONS ANSWERED

Q1-Q2

Nature of AI, intelligence, and classifying real systems

Q3

Farmer's Problem — state space + BFS optimal solution

Q4

Agent, Agent Function, Rationality, Goal-based Agent

Q5

PEAS for Football, Book Shopping, Tennis, Auction

Q6-Q7

Heuristic search vs basic search + state space terms

Q8-Q9

True/False agent assertions + agent function analysis

Q10

9 propositional logic entailments — fully verified

Q11

Agip Estate Club — predicate logic + inference proof

AIMA

Russell & Norvig framework throughout

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